Proofs for the dog game with inertia

This note proves the two propositions of Section 5.20 of the research report. The proofs are mine and have not been refereed; every step is also checked numerically in `tests/test_inertia_theory.py` and the studies `inertia_explicit` and `inertia_random`.

Setting

Rewards are sums over the two coordinates, the dog moves coordinate by coordinate, and the action set is the product of two intervals. So a profile is a feedback equilibrium of the two-dimensional game exactly when it is one in each coordinate: a two-dimensional deviation is a pair of one-coordinate deviations whose payoffs add. Work in one coordinate from now on.

The state is the dog's position x in [0, 1]. Red and blue choose actions a_red and a_blue in [0, 1], and the next position is

f(x, a_red, a_blue) = w0 x + (1 - w0) (w a_red + (1 - w) a_blue),

with inertia w0 in [0, 1), red's weight w in (0, 1) and blue's weight 1 - w. Player i receives -(f - h_i)^2, the squared distance of the next position from its house h_i, discounted by g in [0, 1). A stationary Markov profile pi maps positions to actions, and V_i(x) is player i's discounted payoff from x under pi. The profile is a feedback (Markov-perfect) equilibrium if, at every x, each player's action maximizes -(f - h_i)^2 + g V_i(f) given the other's action. By the one-shot deviation principle (Fudenberg and Tirole, 1991), which applies because payoffs are bounded and discounted, this is the same as having no profitable one-round deviation at any state.

Let (a_red*, a_blue*) be the one-shot Nash actions and d* = w a_red* + (1 - w) a_blue* the one-shot dog. Define

k = (1 - g w0^2) / (1 - g w0), T_i = d* + k (h_i - d*), z_h(x) = w0 x + (1 - w0) d*.

k is at least 1 because w0^2 <= w0. T_i is player i's target and z_h(x) is the next position when both hold.

Lemma 1 (the value of holding)

If both players always play the one-shot actions, then z_h(x) - d* = w0 (x - d*), so the position after t + 1 rounds is d* + w0^(t+1) (x - d*), and summing the geometric series gives, with u = x - d* and e = d* - h_i,

V_i(x) = -[e^2 / (1 - g) + 2 e w0 u / (1 - g w0) + w0^2 u^2 / (1 - g w0^2)].

Lemma 2 (the one-round objective under holding)

If player i plays something else for one round and both hold afterwards, its payoff as a function of the next position z is G_i(z) = -(z - h_i)^2 + g V_i(z). Its second derivative is -2 - 2 g w0^2 / (1 - g w0^2) < 0, so it is strictly concave, and

G_i'(z) = -2 [(z - d*) / (1 - g w0^2) + (d* - h_i) / (1 - g w0)],

which is zero exactly at z = T_i. So G_i increases up to T_i and decreases after it. With the other player at its one-shot action, player i reaches the next positions in an interval of length (1 - w0) w_i (w_red = w, w_blue = 1 - w), which is increasing in its action, and holding gives z_h(x).

Proposition 1

Holding the one-shot actions at every position is a feedback equilibrium if and only if each player at the top of the box ($a_i^* = 1$) has $k(h_i - d^*) \geq w_0(1 - d^*)$, each player at the bottom ($a_i^* = 0$) has $k(d^* - h_i) \geq w_0 d^*$, and, when $w_0 > 0$, no one-shot action is strictly inside the box.

Proof. By Lemma 2 and the one-shot deviation principle, holding is optimal for player i at x exactly when $z_h(x)$ maximizes the strictly concave G_i over i's reachable interval. For a player at the top, $z_h(x)$ is the right end of that interval, so it is optimal iff $G_i'(z_h(x)) \geq 0$, that is $z_h(x) \leq T_i$, that is

$$w_0(x - d^*) \leq k(h_i - d^*).$$

The left side is largest at $x = 1$, which gives the stated condition (for $w_0 = 0$ it always holds, since a one-shot best response at the top has $h_i \geq d^*$). A player at the bottom is the mirror image, with the worst case at $x = 0$. For a player strictly inside, $z_h(x)$ is inside its interval, so holding is optimal iff $z_h(x) = T_i$. A one-shot best response strictly inside the box puts the dog on the player's house, $h_i = d^*$, so $T_i = d^*$, and the condition becomes $w_0(x - d^*) = 0$, which fails at every x other than d^* when $w_0 > 0$. Optimality at every x for every player is the stated set of conditions. ■

Dividing by $1 - d^*$ (or by d^*), the conditions say that the player's slack $s = (h_i - d^*)/(1 - d^*)$ (or $(d^* - h_i)/d^*$) is at least $\rho(w_0, g) = w_0(1 - g w_0)/(1 - g w_0^2)$. ρ rises from 0 to 1 as w_0 goes from 0 to 1 (its derivative in w_0 is $(1 - g w_0(2 - w_0))/(1 - g w_0^2)^2$, at least $(1 - g)/(1 - g w_0^2)^2 > 0$), and its derivative in g is $w_0(w_0^2 - w_0)/(1 - g w_0^2)^2 \leq 0$, so more patient players tolerate more inertia.

Proposition 2

Suppose the coordinate is straddling: after relabeling, $h_{blue} \leq w \leq h_{red}$, so the one-shot actions are $a_{red}^* = 1$, $a_{blue}^* = 0$ and $d^* = w$. Then the policy

$$a_{red}(x) = \text{clip}(1 + (T_{red} - z_h(x))/((1 - w_0) w), 0, 1), \quad a_{blue}(x) = \text{clip}(0 + (T_{blue} - z_h(x))/((1 - w_0)(1 - w)), 0, 1)$$

is a feedback equilibrium for every w_0 in $[0, 1)$ and g in $[0, 1)$.

In words, each player aims the next position at its target, as far as its weight allows, as if the other held its one-shot action.

Proof. If $w_0 = 0$ then $z_h = d^*$, both clips return the one-shot actions, and the game is the one-shot game in every round, so assume $w_0 > 0$. Note $T_{blue} \leq d^* \leq T_{red}$, because $k \geq 1$ and $h_{blue} \leq d^* \leq h_{red}$.

Regions. Let $X_{red} = d^* + (T_{red} - d^*)/w_0$ and $X_{blue} = d^* - (d^* - T_{blue})/w_0$, so that $z_h(X_{red}) = T_{red}$ and $z_h(X_{blue}) = T_{blue}$. Red eases off its edge exactly on $R = \{x : z_h(x) > T_{red}\} = (X_{red}, 1]$, blue exactly on $B = [0, X_{blue})$, and both hold on $H = [X_{blue}, X_{red}]$ intersected with $[0, 1]$. R and B do not meet, because $X_{blue} \leq d^* \leq X_{red}$, so at most one player leaves its edge at any x. The next position under the policy is

$$N(x) = z_h(x) \text{ on } H, \quad \max(T_{red}, z_h(x) - (1 - w_0) w) \text{ on } R, \quad \min(T_{blue}, z_h(x) + (1 - w_0)(1 - w)) \text{ on } B.$$

N is nondecreasing; on R, $T_{red} \leq N(x) < x$ (since $x > X_{red} \geq T_{red}$ and $z_h(x) < x$ when $x > d^*$); on B, $x < N(x) \leq T_{blue}$. H is invariant: for x in H, $N(x) = z_h(x)$ lies between x and d^*, both in H. The targets lie in H when they lie in $[0, 1]$: $T_{red} \leq X_{red}$ because $T_{red} - d^* \geq 0$ and $w_0 \leq 1$, and $T_{red} \geq d^* \geq X_{blue}$; similarly for T_{blue}.

Values. Because H is invariant and both hold on it, V_i equals the holding value of Lemma 1 on H. Let $g_i(z) = -(z - h_i)^2 + g V_i(z)$ be player i's one-round objective as a function of the next position. On H it is G_i of Lemma 2. At any z the policy moves the dog to $N(z)$, so $V_i(z) = g_i(N(z))$, and therefore

g_i(z) = -(z - h_i)^2 + g g_i(N(z)) for every z. (*)

Claim: g_red is nondecreasing on [0, T_red] and nonincreasing on [T_red, 1] (each part when nonempty).

Above the target. Let M be the set of bounded functions phi on [T_red, 1] that equal G_red on [T_red, X_red] and are nonincreasing. Define (A phi)(z) = G_red(z) on [T_red, X_red] and (A phi)(z) = -(z - h_red)^2 + g phi(N(z)) on R; this is well defined because N maps R into [T_red, 1). A is a contraction with modulus g in the sup norm, and by (*) the restriction of g_red to [T_red, 1] is its fixed point. A maps M into M:

- on R, -(z - h_red)^2 decreases, because z > X_red >= T_red >= h_red (T_red - h_red = (k - 1)(h_red - d*) >= 0), and phi(N(z)) does not increase, because N does not decrease and phi does not increase;
- across the boundary, for z1 in [T_red, X_red] and z2 in R, (A phi)(z2) <= -(z2 - h_red)^2 + g phi(T_red) < -(X_red - h_red)^2 + g G_red(T_red) = G_red(X_red) <= G_red(z1), using N(z2) >= T_red, phi(T_red) = G_red(T_red), the identity G_red(X_red) = -(X_red - h_red)^2 + g G_red(z_h(X_red)) with z_h(X_red) = T_red (this is (*) on H), and that G_red decreases after T_red.

M is closed under uniform limits and contains, for example, G_red on [T_red, X_red] extended by the constant G_red(X_red), so iterating A from it converges to the fixed point, which therefore lies in M (Banach's fixed point theorem).

Below the target. The same argument on [0, T_red] with the set of functions that equal G_red on [X_blue, T_red] and are nondecreasing: on B, -(z - h_red)^2 increases because z < X_blue <= d* <= h_red, and N maps B into (z, T_blue], which lies in [0, T_red]; across the boundary, (A phi)(z1) <= -(z1 - h_red)^2 + g G_red(T_blue) < -(X_blue - h_red)^2 + g G_red(T_blue) = G_red(X_blue) <= G_red(z2) for z1 in B and z2 in [X_blue, T_red], using z_h(X_blue) = T_blue, T_blue in [X_blue, T_red] and that G_red increases up to T_red. This proves the claim; the claim for blue is the mirror image, with its peak at T_blue.

Best replies. A function that does not decrease up to T and does not increase after it is maximized over any interval [L, U] at clip(T, L, U). Given blue's action at x, red reaches the next positions [U - (1 - w0) w, U], where U is the position with a_red = 1. On H and R, blue is at 0, so U = z_h(x) and red's best reply is clip(T_red, U - (1 - w0) w, U), which is exactly the policy (it holds on H, where z_h(x) <= T_red, and moves to max(T_red, z_h(x) - (1 - w0) w) on R). On B, blue moves the position up to N(x) <= T_blue <= T_red with red at 1, so U = N(x) <= T_red and the best reply is red's edge, as in the policy. Blue's case is the mirror image. By the one-shot deviation principle the policy is a feedback equilibrium. ■

Many houses

With N players, weights w_i summing to 1 and the dog at sum_i w_i a_i, Lemmas 1 and 2 hold unchanged for each player (the other players' held actions only enter through d*, and player i reaches an interval of length (1 - w0) w_i). So Proposition 1 holds verbatim: holding the one-shot actions is a feedback equilibrium iff every player at an edge has slack at least rho(w0, g) and, when w0 > 0, no player (the pivot) is strictly inside the box. In the one-shot equilibrium, which is the generalized median of Renault and Trannoy (2005), at most one player is inside. The study `many_houses` measures how often there is none, and how the largest admissible inertia scales with N. The target form also carries over numerically: if every player aims at a target, each round is the one-shot game with the targets as houses, and targets from a fixed point give a feedback equilibrium whenever every player's one-round objective is single-peaked (study `many_targets`); there is no proof for many houses.

Continuous-time limit

With time step dt, w0 = 1 - dt and g = exp(-r dt), 1 - g w0 = (r + 1) dt + O(dt^2) and 1 - g w0^2 = (r + 2) dt + O(dt^2), so k -> (r + 2) / (r + 1) and rho -> (r + 1) / (r + 2). In Proposition 2, (T_i - z_h(x)) / ((1 - w0) w_i) = (T_i - x + dt (x - d*)) / (dt w_i), which tends to +infinity where x < T_i and to -infinity where x > T_i, so the policy tends to the bang-bang rule "push the edge toward T_i", with the interior band of width O(dt). This describes the limit of the discrete-time equilibria; that it is an equilibrium of the continuous-time game is not proved here.

Overshooting

In Proposition 2 a player aims the next dog position at T_i = d* + k (h_i - d*), so it aims past its own house by (k - 1)(h_i - d*). Since k = (1 - g w0^2) / (1 - g w0), dk/dw0 has the sign of g w0^2 - 2 w0 + 1, which vanishes at w0 = (1 - s) / g = 1 / (1 + s) with s = sqrt(1 - g). So k = 1 at w0 = 0 and at w0 = 1, and the overshoot is largest at the intermediate inertia 1 / (1 + s), where k = 2 / (1 + s): at most 2, approached by patient players.

Persistence and the median rule

Now N players (N odd) with distinct houses h_1 < ... < h_N report points r_i in [0, 1], and the outcome moves to z = w0 x + (1 - w0) median(r); losses are (z - h_i)^2, discounted. Let h_m be the median house.

Proposition 3. For w0 = 0 truthful reporting (r_i = h_i at every outcome) is a feedback equilibrium; for any w0 in (0, 1) it is not.

Proof. With w0 = 0 the round does not depend on the outcome, so each round is the one-shot median game, where truthful reporting is dominant (Moulin, 1980). With w0 > 0 and everyone truthful, the outcome moves to w0 x + (1 - w0) h_m and converges to h_m, so by Lemma 2 (with d* = h_m, the median player's house) the median player's one-round objective is a concave quadratic in the next outcome peaked at T_m = h_m. By changing its report within [h_(m-1), h_(m+1)] the median player moves the next outcome over an interval that contains w0 x + (1 - w0) h_m strictly inside, and that contains h_m itself whenever x is close enough to h_m. At such an x other than h_m, reporting (h_m - w0 x) / (1 - w0), not h_m, puts the outcome on its peak, a strict gain. ■

Proposition 9 below restores truthful reporting with a receiver that compensates for its inertia.

Proposition 4. For every w0 in [0, 1) the following profile is a feedback equilibrium: the median player reports clip((h_m - w0 x) / (1 - w0), 0, 1), players with houses above h_m report 1 and players below report 0. Under it the outcome moves to G(x) = clip(h_m, w0 x, w0 x + 1 - w0) each round: as close to the median house as can be reached, the path a planner minimising (z - h_m)^2 would choose.

Proof. The median of the reports is the median player's report, which gives G. By the one-shot deviation principle it suffices to check one-round deviations.

- The median player can move the median, hence the next outcome, anywhere in [w0 x, w0 x + 1 - w0]. Its problem is then a one-player control problem with loss (z - h_m)^2: if h_m is reachable it stays reachable forever after, and otherwise the reachable interval moves monotonically with the outcome, so moving as close to h_m as possible each round is optimal.
- A player with house h_i > h_m can only lower the median (by reporting below the median player's report), never raise it. If G(x) <= h_m, lowering the next outcome brings it further from h_i and, since G is nondecreasing, lowers the whole future path, which stays at or below h_m < h_i: a loss. If G(x) > h_m, the median player is reporting 0, so G(x) = w0 x is already the lowest reachable outcome and nothing can be lowered.
- Players below the median are the mirror image. ■

So inertia costs the median rule its truthfulness but not its speed: the outcome still moves to the median house as fast as any mechanism could move it there. The mean rule has no such property; in the study `mechanisms` its outcome moves more slowly, because players on the far side of the outcome keep pulling against it.

Cooling off

Add a shock to each round: the outcome lands at y = clip(z + e, 0, 1), where z is the outcome the reports aim at and e has a symmetric distribution with e != 0 with positive probability. The shock either comes with the new aggregate, e = (1 - w0) s eps (the inertia filters it), or hits the outcome itself, e = s eps. The greedy planner aims at the mean house hbar of the N houses each round, as close as it can reach.

Proposition 5. Let wbar = min(hbar, 1 - hbar). For w0 in [0, wbar] the greedy planner's expected discounted summed loss, from any start, is

L(w0) = (sum_i (h_i - hbar)^2 + N phi(sigma)) / (1 - g), phi(sigma) = E[(clip(hbar + sigma eps, 0, 1) - hbar)^2],

with sigma = (1 - w0) s when the shock comes with the aggregate and sigma = s when it hits the outcome. So with a filtered shock L is strictly decreasing on [0, wbar]: some inertia is strictly better than none, and the welfare-maximizing inertia is at least wbar. With a shock on the outcome, L is constant on [0, wbar].

Proof. From any y in [0, 1] the reachable interval [w0 y, w0 y + 1 - w0] contains hbar when w0 <= hbar and hbar <= 1 - w0, so the planner aims at hbar in every round and every round's outcome is clip(hbar + e, 0, 1). The summed loss at an outcome y is sum_i (h_i - hbar)^2 + N (y - hbar)^2, which gives L. Each |clip(hbar + sigma eps, 0, 1) - hbar| is nondecreasing in sigma and strictly increasing where it is not clipped, which has positive probability, so phi is strictly increasing. ■

Above wbar the planner can no longer reach hbar from every outcome, and slow adjustment starts to cost; the study `mechanisms_noise` finds the loss U-shaped in w0, lowest at w0 = 0.6 to 0.75 for the planner and at 0.45 to 0.9 for the equilibria of the mean and the median rule (which the proposition does not cover), the optimum rising with the size of the shock. With a shock on the outcome the planner is best off with no inertia. (The greedy planner is not optimal when clipping is frequent; the study also computes the optimal planner.)

The receiver as a mechanism

In the Battling Influencers Game (Wu, Zhu, Cai and Zhu, 2025) the receiver averages the influencers' actions, and at every pure Nash equilibrium all influencers but at most one exaggerate to the boundary, the remaining one putting the receiver on its own target (their Theorem 4). The paper's conclusion asks for mechanism design that removes this incentive. With inertia the question becomes: which receiver should aggregate the influencers' actions round after round?

On one coordinate with action space [0, 1] and equal weights, a natural family of receivers is the generalized medians of Moulin (1980): the receiver moves to z = w0 x + (1 - w0) a, where a is the median of the N actions and N - 1 fixed phantoms alpha_1 <= ... <= alpha_{N-1}. In one shot these are exactly the anonymous strategy-proof rules for single-peaked preferences (with Pareto efficiency), and they include the median receiver (phantoms (N - 1)/2 at 0 and (N - 1)/2 at 1).

Proposition 7. With equal weights, the averaging receiver of BIG and the generalized median receiver with phantoms alpha_k = k/N have the same equilibria in target form, for every inertia w0 and discount g, and hence the same paths and the same losses.

Proof. In target form each influencer aims the next position at its target T_i; let u_i = (T_i - w0 x) / (1 - w0) be the aggregate that would put the receiver there. Under averaging, the round is BIG with targets u_i (Section "Many houses"): influencers with u_i > a play 1, those with u_i < a play 0, and at most one with u_i = a plays

inside, so with A = #{u_i > a} and B = #{u_i < a}, A / N <= a <= (N - B) / N. Among the 2N - 1 values {u_i clipped to [0, 1]} and {k / N}, the phantoms above a number at most N - 1 - A (they are k / N > a >= A / N, so k >= A + 1), so at most N - 1 values lie above a; symmetrically at most N - 1 lie below. So a is their median, which is what the generalized median receiver produces when every influencer reports its u_i, a dominant action because its one-round objective is single-peaked in the next position (the assumption of target form). The map from targets to next position is the same for both receivers, so are the fixed-point conditions for the targets. ■

For w0 = 0 this is the static statement that the average voting rule's equilibrium outcome is a generalized median (Renault and Trannoy, 2005), and BIG's Theorem 4 is its form for the actions. Numerically, the two receivers' targets and losses agree to machine precision, and the generalized median equilibria pass the one-round deviation check (test `test_the_mean_rule_is_the_generalized_median_with_evenly_spread_phantoms_and_the_median_rule_with_edge_phantoms`).

Corollary (the phantoms are the speed). While no influencer is pivotal, every influencer plays an edge, and with A of them pushing up the aggregate is the A-th smallest phantom (alpha_0 = 0, alpha_N = 1): the N - A zeros fill the bottom of the 2N - 1 values and the median is the next one. So z = w0 x + (1 - w0) alpha_A. The phantoms fix both where the receiver settles (the generalized median of the targets) and how fast it gets there: averaging moves with drift A / N - x, a tug of war, and the median receiver at full speed, 0 or 1, as soon as a majority pushes one way. Choosing a receiver means choosing this trade-off, which is what the study `mechanisms_phantoms` optimizes.

Without inertia the trade-off has a known answer: for squared distance and targets drawn uniformly, the uniform median (phantoms k/N) is the welfare-maximizing strategy-proof rule (Jennings, Laraki, Puppe and Varloot, 2021). With three influencers and phantoms (p, 1 - p) the expected summed loss is (1 - 10p^3 + 30p^4 - 18p^5)/5, whose derivative -6p^2(3p - 1)(p - 1) makes p = 1/3 the unique minimizer on [0, 1/2] (5/27 against 1/5 for the median receiver). So in the static game BIG's averaging receiver, for all its exaggeration, is the best strategy-proof receiver.

Proposition 8 (slow receivers should use an order statistic). Take equal weights, targets drawn independently from a distribution F on [0, 1], the start x uniform on [0, 1], and a fixed discount g. For a generalized median receiver with phantoms alpha (any monotone alpha, not only symmetric ones), the expected discounted summed loss is

L(alpha; w0) = L_0(w0) + (1 - w0) (2N / (1 - g)^2) (sum_{k=1}^{N-1} c_k alpha_k + c_N) + O((1 - w0)^2),

where L_0(w0) does not depend on alpha and c_k = E[(x - hbar) 1{k targets lie above x}]. The first-order term is minimized by an order-statistic receiver: the one that moves toward the j-th smallest target (phantoms: N - j at 0, j - 1 at 1), with j minimizing E[(h_(j) - hbar)^2], the mean squared distance between that order statistic and the mean target. For uniform targets c_k = (N - 2k)/(2N(N + 2)) and, for N odd, j is the median: as w0 tends to 1 the welfare-best generalized median receiver tends to the median receiver, whereas at w0 = 0 it is BIG's averaging receiver (above). For strongly skewed targets j can lie above the median (for Beta(0.2, 5) with N = 5, j = 4; test `test_the_best_slow_receiver_is_the_order_statistic_closest_to_the_mean_target`).

Proof (first-order argument).

1. *Influencers sit at edges.* As w0 -> 1 the position barely moves within the discount horizon, so an influencer's one-round objective tends uniformly to -(z - h_i)^2 / (1 - g) and its target T_i to h_i. The aggregate it would need, u_i = (T_i - w0 x)/(1 - w0), then diverges unless x = T_i, so every influencer reports 0 or 1 and, by the corollary, the aggregate is alpha_{A(x)} with A(x) the number of targets above x. Starts within O(1 - w0) of a target change this on a set of probability O(1 - w0), which only affects the O((1 - w0)^2) term.

2. *Expansion.* With e = 1 - w0, x_{k+1} = x_k + e (alpha_{A(x_k)} - x_k), so x_k = x + k e (alpha_{A(x)} - x) + O(k^2 e^2), and with l(x) = sum_i (x - h_i)^2, sum_k g^k l(x_{k+1}) = l(x) / (1 - g) + e l'(x) (alpha_{A(x)} - x) / (1 - g)^2 + O(e^2), using sum_k (k + 1) g^k = 1 / (1 - g)^2. Since l'(x) = 2N (x - hbar), the part that depends on alpha is e (2N / (1 - g)^2) E[(x - hbar) alpha_{A(x)}] = e (2N / (1 - g)^2) sum_k c_k alpha_k.
3. *The minimizer.* The objective is linear in alpha over the monotone set 0 <= alpha_1 <= ... <= alpha_{N-1} <= 1, whose extreme points are the thresholds alpha_k = 1{k >= m}, with value sum_{k >= m} c_k = E[(x - hbar) 1{A(x) >= m}]. At least m targets lie above x exactly when x is below q = h_(N-m+1), so integrating over the uniform start, the value is E[integral_0^q (x - hbar) dx] = E[q^2/2 - q hbar] = E[(q - hbar)^2]/2 - E[hbar^2]/2. So the best threshold is the order statistic closest to hbar in mean square.
4. *Uniform targets.* Given x and A = k, the k targets above are uniform on (x, 1) and the others on (0, x), so E[hbar | x, A = k] = x/2 + k/(2N). Given A = k, x has the Beta(N - k + 1, k + 1) density with mean (N - k + 1)/(N + 2), and P(A = k) = 1/(N + 1). So c_k = (1/(N + 1)) ((N - k + 1)/(2(N + 2)) - k/(2N)) = (N - 2k) / (2N(N + 2)), positive for k < N/2 and negative for k > N/2 (never 0 for N odd), and the tail sum is smallest exactly at m = (N + 1)/2, the median. ■

The remainder in step 2 is bounded for each fixed alpha but the argument does not make it uniform over alpha, and the regime needs 1 - w0 small against 1 - g, not just small. Numerically (study `mechanisms_limit`, three influencers, discount 0.9), at w0 = 0.99 and 0.995 the loss rises with p from the median receiver p = 0, with slopes close to the first-order prediction (table `mechanisms limit`). Some of these solves stop at the 600-iteration cap (23 and 48 of 280); on the draws without any capped solve (27 and 22 of 40) p = 0 is still best and the loss still rises with p throughout. With skewed targets the full dynamic game agrees in direction (table `skewed paired`, paired by draw): with 7 influencers and Beta(0.1, 1) targets at w0 = 0.99 the predicted receiver, the 5th smallest target, loses less than the median receiver in 46 of 60 draws, by 0.0080 on average (standard error 0.0025; 0.0095 with s.e. 0.0034 on the 43 draws without a capped solve), about a third of the first-order prediction 0.026, as expected when 1 - w0 is not yet small against 1 - g. With 5 influencers and Beta(0.2, 5) targets the predicted advantage is only 0.001 and the 40-draw test cannot resolve it (the median is marginally ahead, within one to two standard errors). Together with the static result, the welfare-best strategy-proof receiver runs from BIG's averaging receiver without inertia to the median receiver with slow adjustment; the study `mechanisms_phantoms` traces the path in between.

Compensating for inertia

Proposition 3 shows that persistence breaks the median receiver's strategy-proofness: the receiver moves only part of the way toward the median report, so the median player exaggerates to make up for it. A receiver that compensates for its own inertia restores it. The *compensated* receiver with phantoms alpha moves to

z = clip(m(r), w0 x, w0 x + 1 - w0), m(r) = the median of r_1, ..., r_N, alpha_1, ..., alpha_{N-1},

the reachable outcome closest to the generalized median of the reports (`mechanisms.compensated`). With w0 = 0 it is the generalized median itself.

Proposition 9 (compensated receivers are strategy-proof under inertia). For any phantoms alpha, any number of players, every w0 in [0, 1) and every discount g, truthful reporting (r_i = h_i at every outcome) is a feedback equilibrium of the compensated receiver. Along it the outcome moves each round as close as it can to s = m(h); for the median receiver this is the outcome path of the equilibrium in Proposition 4.

Proof.

1. Under truthful reporting the outcome follows G(x) = clip(s, w0 x, w0 x + 1 - w0). G is nondecreasing in x, moves x toward s without crossing it, and fixes s. So paths from two starting points keep their order in every round, and a path that starts at or below s stays at or below s (and likewise above).

2. Take a player with h_i > s at outcome x, and let [a, b] = [w0 x, w0 x + 1 - w0]. Its truthful report is above s; moving it anywhere in [s, 1] leaves the generalized median at s, and moving it to r < s gives a median m' in [r, s]. So any deviation gives z' = clip(m', a, b) <= z = clip(s, a, b). - If z > s, then a > s >= m', so z' = a = z and nothing changes. - If z <= s, then z' <= z <= s, and by step 1 the path from z' lies weakly below the path from z in every round, both at or below s < h_i. Every round's loss is weakly larger after the deviation.
3. A player with h_i < s is the mirror image. For a player with h_i = s a deviation either leaves z unchanged or moves it away from s, and by step 1 every later outcome is then at least as far from s.
4. No one-round deviation gains at any outcome, so by the one-shot deviation principle (discounting, bounded losses) truthful reporting is a feedback equilibrium. ■

Corollary (any strategy-proof rule). The proof uses only one property of the generalized median m: a player whose target is above m(h) cannot raise it by changing its report (and the mirror image). Every generalized median voter scheme has this property, anonymous or not (for instance a dictator, or max(min(r_1, 0.7), min(r_2, r_3))); by Moulin (1980) and Border and Jordan (1983) these are the rules that depend only on the reports and are strategy-proof for single-peaked preferences. So compensating for inertia keeps every such rule truthful at every inertia. It does not make a manipulable rule truthful: with compensated averaging of the reports, clip(mean(r), w0 x, w0 x + 1 - w0), truthful reporting is not an equilibrium (it is the one-shot averaging game at w0 = 0; numerically also at w0 = 0.6, test `test_compensation_keeps_every_strategy_proof_rule_truthful_but_not_averaging`).

The proof never uses the squared loss: any per-round loss that increases with the distance to h_i will do. Compensation changes the outcome for receivers other than the median one. Every compensated receiver moves at full speed, so the speed advantage that makes the median receiver best at high inertia (Proposition 4 and the study `mechanisms_phantoms`) disappears, and only the endpoint s matters.

Numerically (study `mechanisms_compensated`, table `mechanisms compensated`, on the 100 draws of `mechanisms_wide` for 3, 5 and 7 players at seven inertias from 0 to 0.9, discount 0.9):

- No one-round deviation from truthful reporting gains anything, for the compensated averaging (phantoms k/N) and median receivers, on the first 3 draws of every cell (largest gain 0 on a 101-point report grid; tests `test_truthful_reporting_is_an_equilibrium_once_the_receiver_compensates_for_inertia` and `test_the_compensated_median_with_truthful_reports_follows_the_median_rule_equilibrium`). Without compensation the same profile loses to a deviation (Proposition 3).
- The compensated median receiver's loss equals the plain median rule's equilibrium loss to within the grid error of the equilibrium solver (at most 0.015).
- With truthful reports the compensated averaging receiver has a lower mean loss than the equilibria of both plain receivers at every positive inertia. At w0 = 0.9 it is below plain averaging by 0.20, 0.33 and 0.52 (3, 5 and 7 players; lower by more than 0.015 in 86, 96 and 100 of 100 draws), and below the median receiver by 0.08, 0.17 and 0.18 (standard errors 0.02 to 0.05). Against the median receiver this holds on average, not draw by draw: at each inertia it is lower in 27 to 53 draws and higher in 14 to 18.
- Among compensated receivers with phantoms (p, 1 - p) for 3 players, the best on a 0.025 grid is p = 0.325, the grid point nearest averaging, at every inertia (numerical; not proved). So the reversal between averaging and the median receiver comes from receivers that do not compensate for their inertia.

The one-round step of the proof is the static fact that moving to the feasible point nearest a median keeps the rule strategy-proof (Walsh 2021, Theorem 3, for facilities at limited locations); what is added here is the dynamic step, the ordering of paths. Proposition 9 gives one equilibrium of the compensated receiver; it does not rule out others.

Proposition 10 (with compensation, slow receivers keep the static optimum). Take equal weights, targets drawn independently from a distribution F on [0, 1], the start x uniform on [0, 1], a fixed discount g and truthful reports. For the compensated receiver with phantoms alpha, with s = m(h) the generalized median of the targets and phantoms,

L(alpha; w0) = L_0(w0) + (1 - w0) (N / (1 - g)^2) E[(s - hbar)^2] + O((1 - w0)^2),

where L_0(w0) does not depend on alpha; at w0 = 0 the loss is (N / (1 - g)) E[(s - hbar)^2] plus a constant. So without inertia and for a slow receiver alike, the welfare-best compensated receiver minimizes E[(s - hbar)^2], the static criterion: for uniform targets, BIG's averaging receiver (Jennings, Laraki, Puppe and Varloot, 2021; p = 1/3 for three players by the closed form above). Without compensation the slow optimum is the median receiver instead (Proposition 8).

Proof (first-order argument, as for Proposition 8). Under truthful reports the compensated receiver moves at full speed toward s and stops there: x_{k+1} = x_k + e (1{s > x_k} - x_k) with e = 1 - w0, until it reaches s. This is step 2 of the proof of Proposition 8 with the aggregate alpha_{A(x)} replaced by 1{s > x}; step 1 is not needed, since the reports are truthful and the receiver does the moving. Starts within O(e) of s, where the stop matters within the discount horizon, have probability O(e) and only affect the O(e^2) term. So L = L_0 + e (2N / (1 - g)^2) E[(x - hbar) 1{s > x}] + O(e^2), and with x uniform, E[(x - hbar) 1{s > x}] = E[s^2 / 2 - s hbar] = (E[(s - hbar)^2] - E[hbar^2]) / 2. ■

The same caveats as for Proposition 8 apply: the remainder is not shown to be uniform in alpha, and the regime needs 1 - w0 small against 1 - g. For three players the first-order prediction matches the loss gap between the median receiver and averaging to within 15% at w0 = 0.995 (test `test_a_slow_compensated_receiver_is_ranked_by_the_static_criterion`; about 4% with 2000 draws).

Across inertias, study `mechanisms_compensated_design` (table `mechanisms compensated design`) searches symmetric phantoms on fine grids, plus exact averaging: (p, 1 - p) in steps of 0.005 for 3 players (1000 draws, discount 0.9 and 0.99), steps of 0.025 for 5 players and 0.05 for 7 (500 draws, discount 0.9), at eight inertias from 0 to 0.995. In every one of the 32 cells the best shape is the same as without inertia: p = 0.325 for 3 players (the grid point next to 1/3, ahead of exact averaging by at most 0.0033, well within one standard error of the paired difference), (0.275, 0.4) for 5 players (within 1.7 standard errors of exact averaging (0.2, 0.4), the static optimum), and exact averaging itself for 7 players. The median receiver loses more than averaging in every cell, by 8.9 to 10.4 standard errors, and the gap shrinks as inertia grows, as the first-order term predicts. So with compensation the welfare-best receiver does not depend on inertia at all in these cases, in contrast with the plain receivers (study `mechanisms_phantoms`). Proposition 13 below shows that this is exact: for uniform targets the optimum is averaging at every inertia, and the grid points that came out ahead of it did so within sampling error.

Receivers that move within a reachable set

Propositions 9 and 10 are about proportional inertia, where the outcome can reach [w0 x, w0 x + 1 - w0] in one round. The arguments only use two properties of that interval, so they hold for a wide class of receivers. Let the receiver's *reachable set* from x be an interval R(x) = [a(x), b(x)] in [0, 1] with

(R1) a(x) <= x <= b(x): the receiver can always stay where it is; (R2) a and b are nondecreasing: starting higher never shrinks what can be reached upward or extends it downward.

Proportional inertia (a = w0 x, b = w0 x + 1 - w0), a speed limit (a = max(0, x - c), b = min(1, x + c)) and asymmetric adjustment (different limits up and down) all satisfy (R1) and (R2). The *compensated* receiver with rule m moves to clip(m(r), a(x), b(x)), the reachable point closest to m(r).

Proposition 11 (compensation keeps every strategy-proof rule truthful). Under (R1) and (R2), for every generalized median voter scheme m, any number of players, any discount and any per-round loss that increases with the distance to the player's target, truthful reporting is a feedback equilibrium of the compensated receiver. Along it the outcome after t rounds is clip(s, A_t, B_t), where s = m(h), A_0 = B_0 = x_0, A_{t+1} = a(A_t) and B_{t+1} = b(B_t), and [A_t, B_t] is the set of outcomes reachable in t rounds.

Proof. The proof of Proposition 9 goes through word for word with G(x) = clip(s, a(x), b(x)). G is nondecreasing (a composition of nondecreasing maps, by R2), moves x toward s without crossing it, and fixes s (both by R1). These are the only facts step 1 used; steps 2 to 4 used only that a deviation by a player with target above s can only lower m (the uncompromising property of Border and Jordan, 1983) and that the clipped outcome then weakly falls. For the path: the set reachable in t + 1 rounds is the union of [a(y), b(y)] over y in [A_t, B_t]; each of these intervals contains its y (R1) and their endpoints are monotone in y (R2), so the union is [a(A_t), b(B_t)]. If s is in [A_t, B_t], the truthful outcome is s and stays there (R1). If s > B_t, the truthful outcome is B_t by induction, and the next one is min(s, b(B_t)) because a(B_t) <= B_t < s. The case s < A_t is the mirror image. ■

Tests `test_compensation_keeps_truthful_reporting_for_any_monotone_reachable_set_that_allows_staying` (a speed limit and asymmetric limits, for the median, averaging and dictator rules) and `test_truthful_reporting_fails_when_the_receiver_cannot_stay_or_its_reach_falls` (the two examples below) check this numerically. Neither condition can be dropped. In both examples there is one player (the rule is a dictator, which is strategy-proof), the loss is squared and the discount is 0.9.

- *Without (R1).* R(x) = [min(1, x + 0.2), min(1, x + 0.4)]: the receiver must move up by at least 0.2. With target 0.5 and start 0, truthful reporting goes 0.4, 0.6, 0.8, 1, ...; reporting 0.3 first goes 0.3, 0.5, 0.7, 0.9, 1, ... and loses 0.085 less in total.
- *Without (R2).* a(x) = x, b(x) = min(1, x + 0.6) for x <= 0.45 and b(x) = x above 0.45. With target 1 and start 0, truthful reporting moves to 0.6 and is stuck there (loss 0.16 per round, 1.6 in total); reporting 0.4 first loses 0.36 once and then reaches 1 for good, a gain of 1.24.

Compensate the outcome, not the phantoms. Proposition 11 clips the rule's outcome into one reachable interval. Slowing each phantom down on its own is not enough, even when each phantom moves monotonically toward a fixed point. Example: two influencers with targets 0.5 and 1, discount 0.9, squared loss, and the rule x' = median(r_1, r_2, 0, min(1, x + 0.3), 1), whose middle phantom heads for 1 at speed 0.3 while the others stay put. From x = 0 the truthful path is 0.5, 0.8, 1, 1, ...; if the first influencer reports 0 once, the path is 0.3, 0.6, 0.9, 1, ..., and its loss falls by 0.081 + 0.2025 - 0.1786 = 0.1049. The drifting phantom keeps pulling the outcome past the first influencer's target, so slowing the outcome down early pays. In random three-influencer games, phantoms that approach their own fixed points at different speeds gave gains up to 0.067. Test `test_slowing_each_phantom_separately_does_not_keep_reporting_truthful` checks the example.

The welfare of a compensated receiver, exactly

Proposition 12 (welfare is a weighted static criterion). Take a receiver satisfying (R1) and (R2), any distribution of the start x_0, truthful reports, and a social loss per round Lambda(x) = sum_i l(x - h_i) with l convex and differentiable; Lambda is minimized at s*. Let

W(u) = E sum_{t >= 1} g^(t-1) 1{A_t <= u <= B_t},

the expected discounted number of rounds in which u is reachable. Then:

1. The planner (any feasible path, minimizing the discounted social loss) follows clip(s*, A_t, B_t): the compensated receiver with end point s*.

2. The discounted social loss of the compensated receiver with end point s is

L(s) = L(s*) + integral from s* to s of Lambda'(u) W(u) du.

For squared loss, s* is the mean target hbar and L(s) - L(s*) = 2N integral_hbar^s (u - hbar) W(u) du, a Bregman divergence of the convex function with second derivative W.

So L is single-dipped in s with its minimum at the planner's end point. Two compensated receivers whose end points lie on the same side of s* are ranked the same way at every inertia, discount and start distribution: the one closer to s* is better. Without inertia W is the constant 1 / (1 - g), and L(s) - L(s*) = N (s - hbar)^2 / (1 - g), the static criterion. The plain median receiver's equilibrium follows the compensated median's path (Proposition 4), so its loss is also L(h_m).

Proof. (1) By Proposition 11 the outcomes reachable at round t are [A_t, B_t] whatever path was taken, and clip(s*, A_t, B_t) minimizes the convex Lambda over that interval; the path clip(s*, A_t, B_t) is itself feasible, so it minimizes the loss in every round at once. (2) L(s) = E sum_t g^(t-1) Lambda(clip(s, A_t, B_t)). The map s -> clip(s, A, B) is Lipschitz with derivative 1{A < s < B}, so dL/ds = Lambda'(s) W(s) for almost every s (dominated convergence: Lambda' is bounded on [0, 1]); integrate from s*. ■

`mechanisms.reach_weight` computes W for proportional inertia and the evenly spread starts of `social_loss`; test `test_a_compensated_receiver_loses_the_planners_loss_plus_a_weighted_static_criterion` checks the formula against simulated losses (to quadrature error, about 1e-6 relative) at inertia 0, 0.6 and 0.95.

Choosing the receiver: persistence does not move the optimum

Proposition 13 (inertia keeps the sign pattern of the design criterion). Let the targets be independent with a continuous distribution F, the loss squared, and the phantoms ordered, alpha_1 < ... < alpha_{N-1}. Under the assumptions of Proposition 12, with W > 0 on (0, 1), for almost every alpha_k (every alpha_k at which W is continuous)

dL/d alpha_k = 2N W(alpha_k) g_k(alpha_k), g_k(a) = E[(a - hbar) 1{exactly N - k targets lie below a}].

g_k depends on alpha_k alone, so L is a sum of one function of each phantom plus a constant, and each term rises and falls on the same intervals as without inertia (W > 0 only rescales the slope). Hence:

1. If g_k changes sign once, from negative to positive, its zero is the unique minimizer of the k-th term at every inertia, discount and start distribution. When this holds for every k (and the zeros are ordered), the welfare-best phantoms are the static ones, whatever the reachable sets. For uniform targets,

dL/d alpha_k = N W(alpha_k) C(N, k) alpha_k^(N-k) (1 - alpha_k)^k (alpha_k - k/N),

so the unique optimum is alpha_k = k/N, BIG's averaging receiver, at every inertia.

1. If g_k changes sign several times, the local minimizers of the k-th term are the same at every inertia, but which of them is the global minimum depends on W, so inertia and the start distribution can move the optimum. Example: N = 2 (one phantom), targets drawn from 0.455 U[0, 0.04] + 0.1 U[0.48, 0.52] + 0.445 U[0.96, 1], discount 0.9, start at 1. The two local minima are near 0.456 and 0.543. Without inertia L(0.5432) - L(0.456) = +7.6e-4, so the best phantom is near 0.456; with proportional inertia 0.97 the difference is -2.1e-4, so the best phantom is near 0.543. The weight W(u) for this start falls toward the bottom of [0, 1], which is reached late, and that tips the balance between the two local minima.

Proof. With ordered phantoms, alpha_k is the median of the 2N - 1 points exactly when N - 1 points lie below it, k - 1 of them phantoms, so exactly N - k targets. On that event moving alpha_k moves s one for one;

otherwise s does not move. By Proposition 12, L depends on alpha only through s, and dL/ds = 2N (s - hbar) W(s); on the event s = alpha_k, so dL/d alpha_k is as stated (differentiation under E is justified because s is Lipschitz in alpha). For uniform targets, with j = N - k targets below a, the targets below have mean a / 2 and those above (1 + a) / 2, so E[(a - hbar) 1{j below}] = C(N, j) a^j (1 - a)^(N-j) (a - (j a / 2 + k (1 + a) / 2) / N) = C(N, k) a^(N-k) (1 - a)^k (a - k/N) / 2, which changes sign once, at k/N. L is Lipschitz in each alpha_k, so it is the integral of this almost-everywhere derivative: a term is decreasing where g_k < 0 and increasing where g_k > 0, whatever W > 0 is. That gives 1 and the first half of 2; the example is computed directly from the compensated paths (test `test_with_clustered_targets_inertia_can_move_the_best_compensated_phantom`). ■

An earlier version of this proposition claimed part 1 for every target distribution; the example above, found in an independent check of the proofs, shows that it needs the single sign change.

Tests `test_moving_a_phantom_changes_the_loss_at_the_static_rate_times_the_weight` (finite differences of the simulated loss, to 0.1%, at inertia 0, 0.6 and 0.9) and `test_for_uniform_targets_the_phantom_condition_has_its_closed_form_and_vanishes_at_k_over_n` check the formulas. The separability needs ordered, distinct phantoms; when the minimizers of the separate terms are not ordered the ordering constraint binds, which uniform targets never meet.

So for uniform targets compensation removes the reversal completely: the averaging receiver is optimal at every inertia, and inertia changes only how much is lost, not which receiver is best. With clustered targets inertia can still switch between the static local optima (part 2), but never create a new one. Without compensation the slow-receiver optimum is the median receiver (Proposition 8). Proposition 13 also explains the numerical design study (`mechanisms_compensated_design`): its best grid shapes differ from k/N only within sampling error.

Inertia never raises a compensated receiver's price of anarchy

Proposition 14 (inertia only helps relative to the planner). Under the assumptions of Proposition 12, with l >= 0 and Lambda(s*) > 0 (for squared loss: not all targets equal), write L_R for the loss under reachable sets R and L_0 for the receiver without inertia (R(x) = [0, 1]). For every end point s,

L_R(s) - L_R(s*) <= L_0(s) - L_0(s*) = (Lambda(s) - Lambda(s*)) / (1 - g) and L_R(s*) >= L_0(s*) = Lambda(s*) / (1 - g),

so the price of anarchy L_R(s) / L_R(s*) is at most its static value Lambda(s) / Lambda(s*). If the reachable sets shrink, R'(x) contained in R(x) for every x with R' also satisfying (R1) and (R2), the price of anarchy can only fall. For proportional inertia it is therefore nonincreasing in the inertia w0, and when the start has a continuous distribution it tends to 1 as w0 -> 1.

Proof. W(u) is a discounted count of rounds, so W <= 1 / (1 - g), the value without inertia. The integrand Lambda'(u) W(u) in Proposition 12 has the sign of (u - s*) along the path of integration (Lambda is convex), so the excess loss is at most the same integral with W replaced by 1 / (1 - g). The planner's position clip(s*, A_t, B_t) is never closer to s* than s* itself, so its loss is at least Lambda(s*) / (1 - g). If R' is contained in R and R satisfies (R2), then by induction A'_t >= A_t and B'_t <= B_t (A'_{t+1} = a'(A'_t) >= a(A'_t) >= a(A_t)), so W' <= W and the planner's position under R' is never closer to s* than under R: the excess falls and the planner's loss rises. Proportional inertia has t-round sets [w0^t x, 1 - w0^t (1 - x)], which shrink as w0 grows. As w0 -> 1 each indicator in W(u) tends to 1{x = u}, which has probability 0, so W -> 0 by dominated convergence and the excess loss vanishes while the planner's loss stays at least Lambda(s*) / (1 - g). ■

The plain median receiver's equilibrium follows the compensated median's path (Proposition 4), so the bound holds for it too. BIG's plain averaging receiver behaves differently: its mean price of anarchy rises with inertia,

from 1.104 to 1.155 for three influencers (study `mechanisms_wide`). Test `test_inertia_never_raises_a_compensated_receivers_price_of_anarchy` checks the bound and the monotonicity on random games.

What the propositions do not cover

- **Interior one-shot actions.** If a player's one-shot action is strictly inside the box, its hold region is the single point d*, and off it that player pushes at an edge; the dog then returns faster than under holding, the continuation in Lemma 2 is wrong, and the other player's best target moves. In one random game (houses 0.754 and 0.538, w = 0.347, w0 = 0.767, g = 0.335) the dynamic program's red aims at about 0.785 where the policy of Proposition 2 aims at 0.771. The policy fails in 53 of 123 such random games. Because a player's one-round objective depends only on the next position, an equilibrium in target form still exists whenever the objectives stay single-peaked; with the targets from a two-dimensional fixed point (`target_equilibrium`), this holds numerically in 122 of the 123 games. A proof is open.
- **Uniqueness.** The propositions give one feedback equilibrium; they do not rule out others.
- **Shocks.** Propositions 9 to 13 assume a deterministic receiver. With a normal shock on each next outcome, clipped to [0, 1], truthful reporting under the compensated median is no longer exactly an equilibrium. With targets 0.15, 0.3 and 0.8, shocks of standard deviation 0.15 and discount 0.9, the median influencer gains 0.0024 by shading its report from 0.3 to 0.255 at outcome 0.125 with inertia 0.8, against 0.0003 without inertia (from clipping alone); with targets 0.2, 0.5 and 0.8 no gain was found (test `test_with_clipped_shocks_the_median_influencer_gains_a_little_by_shading`).
- **Shared coordinates.** When the houses coincide, both one-shot actions can be interior and the one-shot game has a continuum of equilibria.